2D Rectangle Bimodality Report

Two-Target Width–Start Scan and One-Target Corridor Conditions

Executive Summary

This report asks: **when a 2D domain is not $N \times N$ but a horizontal rectangle, under what conditions does the first-passage time (FPT) distribution become visibly double-peaked?**

We study two reproducible settings:

1. **Two targets (absorbing)** at the two ends of the rectangle. We place the start closer to one end, design a short vs. long biased path (two timescales), and scan **rectangle width** W_y and **start position** x_0 to build a phase map (Fig. 2).
2. **One target with a reflecting-wall corridor**. Start and target sit near the two ends; reflecting walls form a corridor with local bias inside. We compare **with/without global bias** and scan W_y (Fig. 14).

Mechanistically, both constructions share the same template: the total FPT is the sum of two channel components; when the channel modes are separated and the weights are not extreme, the total curve shows two local maxima.

1 Model Definition

Rectangular domain

$$\Omega = \{0, \dots, L_x - 1\} \times \{0, \dots, W_y - 1\}.$$

Base lazy random walk For interior sites,

$$P((x, y) \to (x \pm 1, y)) = \frac{q}{4}, \tag{1}$$

$$P((x, y) \to (x, y \pm 1)) = \frac{q}{4}, \tag{2}$$

$$P((x, y) \to (x, y)) = 1 - q. \tag{3}$$

Reflecting boundary means out-of-domain probability mass is added back to stay probability.

Local-bias rule At biased sites, shift a fraction δ of stay probability to one arrow direction:

$$\Delta = \delta\, p_{\text{stay}}, \quad p_{\text{stay}} \leftarrow p_{\text{stay}} - \Delta, \quad p_{\text{arrow}} \leftarrow p_{\text{arrow}} + \Delta.$$

Global bias (optional) We adjust directional move probabilities by (b_x, b_y) and renormalize so that the total move mass stays equal to q.

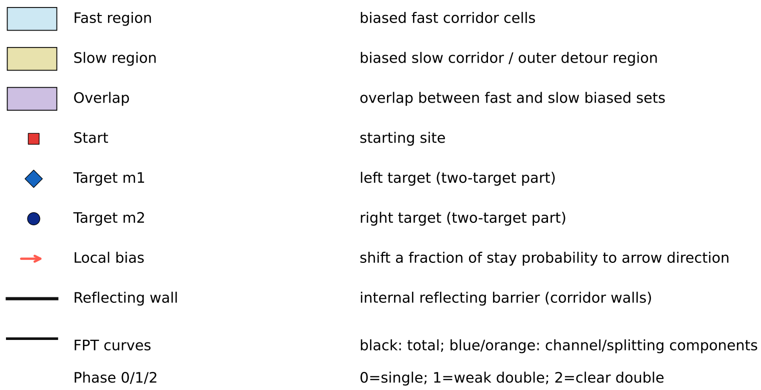

Figure 1: Unified symbol/line/phase legend generated by the same plotting script.

Element	Visual	Meaning (consistent across all figures)
Background		reflecting-boundary rectangle lattice.
Fast region		fast corridor cells.
Slow/outer region		slow corridor or outer-detour region.
Overlap		overlap between biased sets.
Reflecting wall		internal reflecting barrier (corridor wall).
Local bias arrow	$\rightarrow$	shift stay probability to arrow direction.
Heatmap scale		conditional occupancy $P(X_t = n \mid T > t)$.
Total FPT		total FPT pmf.
Component curves		splitting/channel components.
	0	
	1	
Phase 0/1/2	2	0=single; 1=weak double; 2=clear double.

Table 1: Visual legend table.

2 Figure Reading Guide

3 Study I: Two Targets at the Two Ends

3.1 Construction and scan variables

Targets are placed near the left/right ends on the midline:

$$m_1 \approx (1, y_{\rm mid}), \quad m_2 \approx (L_x - 2, y_{\rm mid}), \quad n_0 = (x_0, y_{\rm mid}).$$

We design:
- a short path (fast channel) from n_0 to m_1,
- a long straight path (slow channel) from n_0 to m_2 (longer in the x direction).

By default, we apply local bias only on these two one-cell-thick straight streams, so the arrows do not turn and the number of biased sites is small. We scan (W_y, x_0) and classify outcomes into phase 0/1/2.

3.2 Phase maps and overview

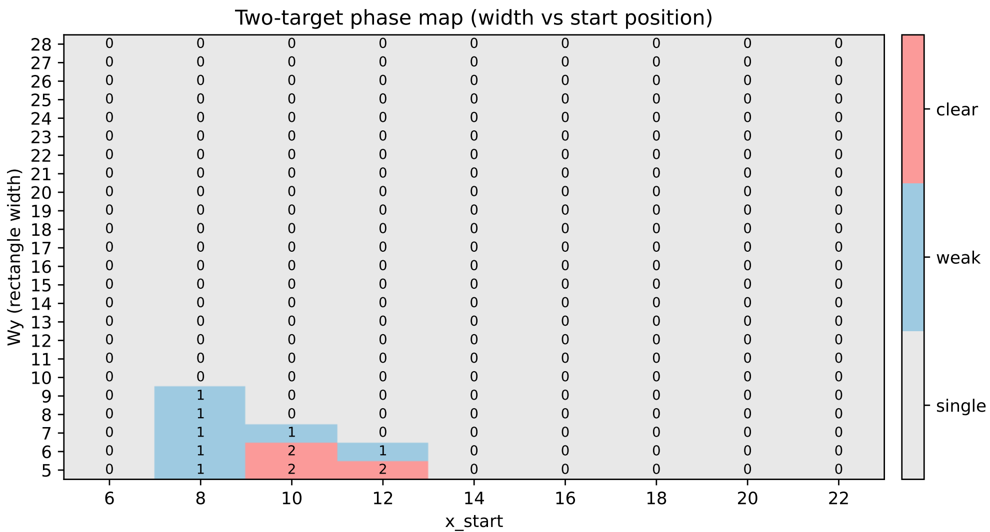

Figure 2: Two-target phase map on (W_y, x_0).

3.3 Representative configurations and double-peak curves

3.4 Width-evolution on a fixed branch ($x_0 = 10$)

To directly answer how bimodality changes with rectangle width, we keep the straight-stream construction fixed and sample one branch ($x_0 = 10$) across transition-focused widths ($W_y \in \{5, 6, 7, 8, 14, 24\}$): the first two are clear-double examples, the third is weak-double (phase 1), and the last three show post-instability single-peak behavior.

Figures 12–13 show that on the fixed branch $x_0 = 10$, clear bimodality is retained at $W_y = 5, 6$, then degrades to weak-double at $W_y = 7$ (phase 1), and fails the robust criterion from $W_y = 8$ onward (phase 0), where wider cases remain single-peaked. Here splitSep is defined on splitting-component mode times and does not by itself imply a double peak in the total curve; phase is assigned by total-curve peak detection together with valley-depth and weight thresholds. Together with Fig. 5 and Table 3, the critical width is branch-dependent: under the current criterion, $x_0 = 8$ has no clear interval; $x_0 = 10$ has clear interval $W_y = 5$–6 (first loss at 7, phase 1); and $x_0 = 12$ has clear interval $W_y = 5$ (first loss at 6, phase 1). Hence there is no single global critical width. Here "critical" means a **width threshold**, not a single critical time;

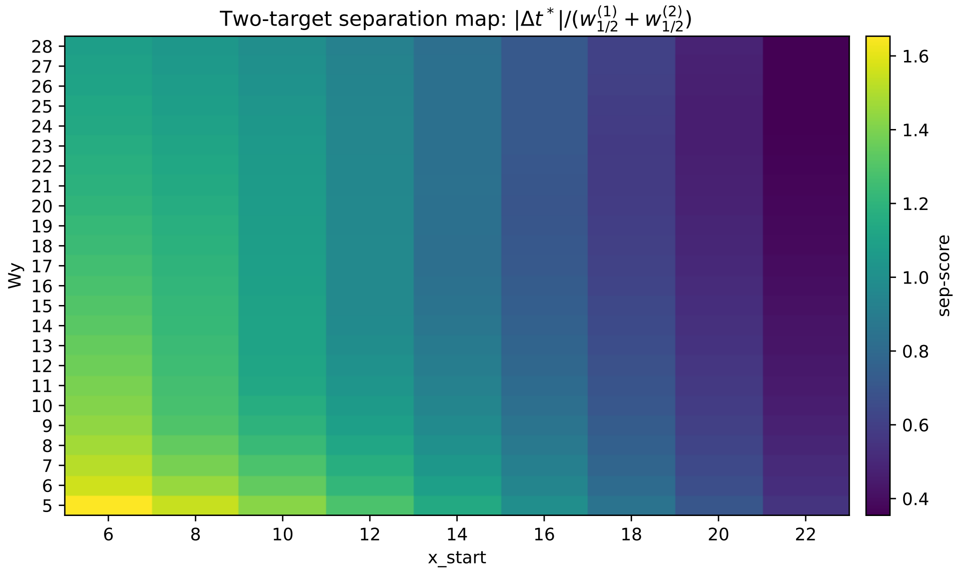

Figure 3: Two-target separation score map $|\Delta t^*|/(w_{1/2}^{(1)} + w_{1/2}^{(2)})$.

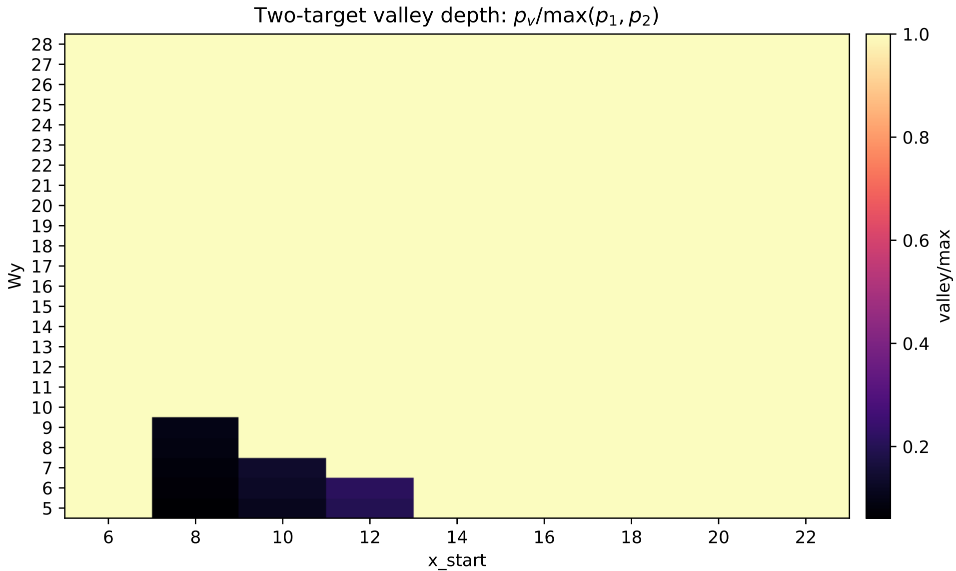

Figure 4: Two-target valley depth map $p_v/\max(p_1, p_2)$.

W_y	n(single)	n(weak)	n(clear)	best $\min(p_1, p_2)$
5	6	0	3	0.456
6	6	0	3	0.444
7	7	0	2	0.411
8	8	0	1	0.374
9	8	0	1	0.364
10	9	0	0	-
11	9	0	0	-
12	9	0	0	-
13	9	0	0	-
14	9	0	0	-
15	9	0	0	-
16	9	0	0	-
17	9	0	0	-
18	9	0	0	-
19	9	0	0	-
20	9	0	0	-
21	9	0	0	-
22	9	0	0	-
23	9	0	0	-
24	9	0	0	-
25	9	0	0	-
26	9	0	0	-
27	9	0	0	-
28	9	0	0	-

Table 2: Two-target width-wise overview of phase counts.

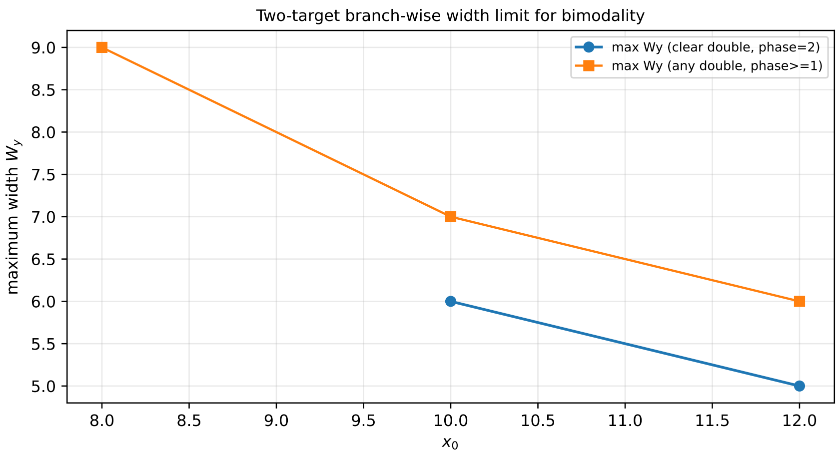

Figure 5: Two-target branch-wise upper width limits (clear and weak/clear), with key branches $x_0 = 8, 10, 12$ extended to $W_y = 60$.

x_0	clear interval(s)	upper clear branch	first loss width	phase at loss
8	-	-	-	-
10	5-6	5-6	7	1
12	5	5	6	1

Table 3: Two-target branch-critical widths for $x_0 = 8, 10, 12$ under the extended width scan (clear intervals and first loss width).

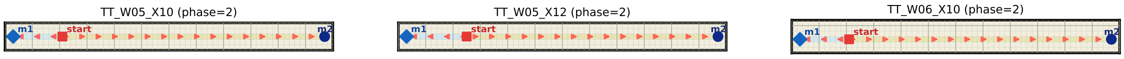

Figure 6: Representative clear-double two-target configurations. The panel now overlays per-cell lattice lines, with a darker reference line every 5 cells, so grid scale and channel length are directly readable.

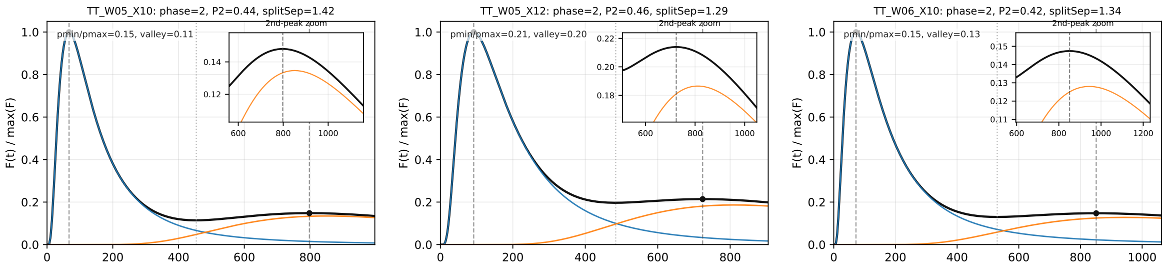

Figure 7: Representative clear-double FPT curves (normalized, peak/valley markers, with a dedicated second-peak zoom inset), aligned with the clear branches selected in this run.

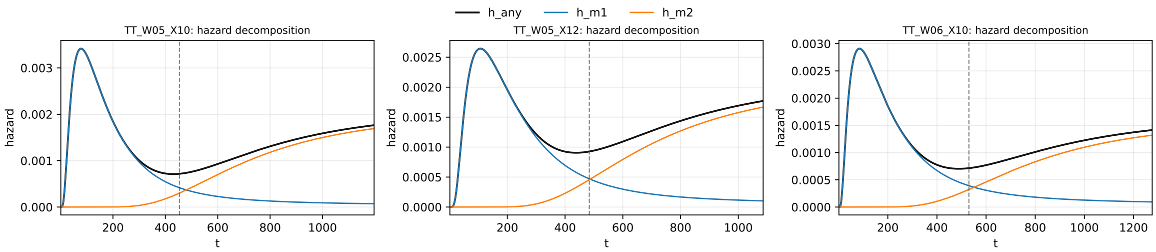

Figure 8: Two-target hazard decomposition ($h = h_1 + h_2$) with x-range adapted to include the second-peak time scale.

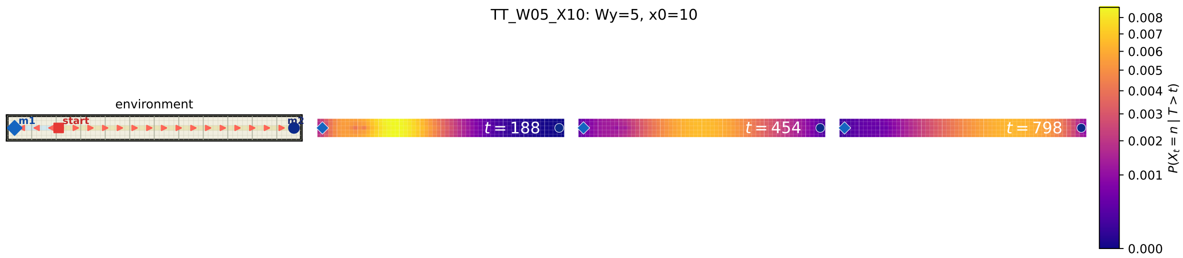

Figure 9: Representative case A: environment + three conditional-occupancy heatmaps (geometry-preserving aspect ratio; no stretching).

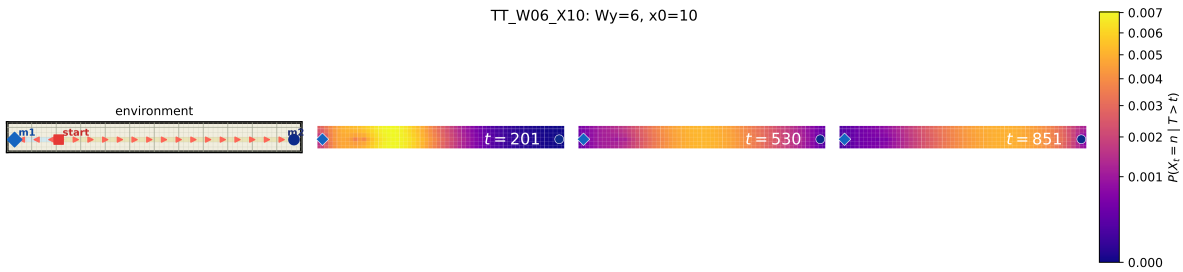

Figure 10: Representative case B: environment + three conditional-occupancy heatmaps (geometry-preserving aspect ratio; no stretching).

ID	W_y	x_0	phase	$P(m_1)$	$P(m_2)$	sep-score
TT_W05_X10	5	10	2	0.561	0.439	1.42
TT_W05_X12	5	12	2	0.544	0.456	1.29
TT_W06_X10	6	10	2	0.575	0.425	1.34

Table 4: Representative two-target metrics (splitting weights, separation score, phase).

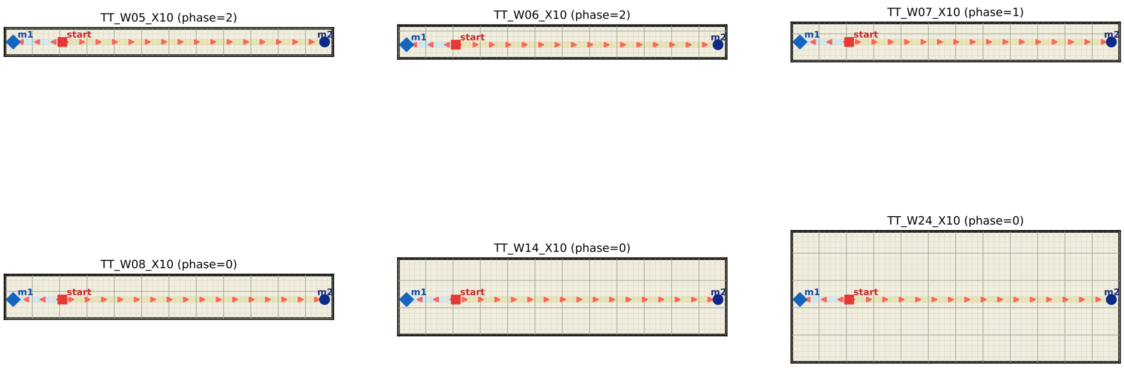

Figure 11: Two-target geometry comparison across widths on the fixed branch $x_0 = 10$.

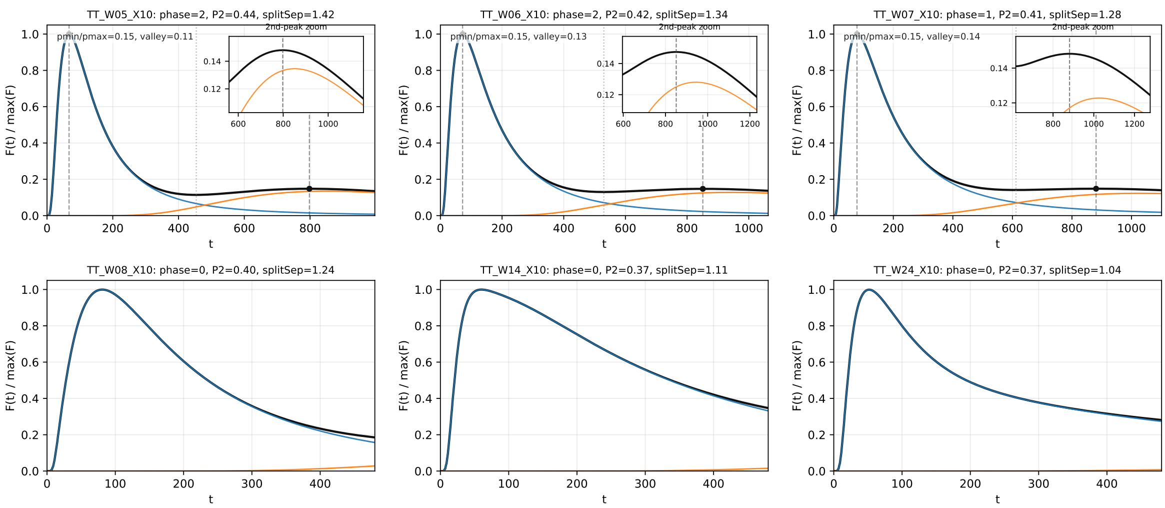

Figure 12: Two-target FPT curves across widths on the fixed branch $x_0 = 10$ (normalized; second-peak zoom inset shown only when a second peak is detected).

ID	W_y	x_0	phase	t_{p1}	t_{p2}	$p_{\min}/p_{\max}$	valley/max
TT_W05_X10	5	10	2	67	798	0.148	0.114
TT_W06_X10	6	10	2	72	851	0.148	0.130
TT_W07_X10	7	10	1	78	881	0.148	0.141
TT_W08_X10	8	10	0	-	-	-	-
TT_W14_X10	14	10	0	-	-	-	-
TT_W24_X10	24	10	0	-	-	-	-

Table 5: Fixed-branch width-sweep metrics (including phase, $t_{p1}, t_{p2}, p_{\min}/p_{\max}$, valley/max).

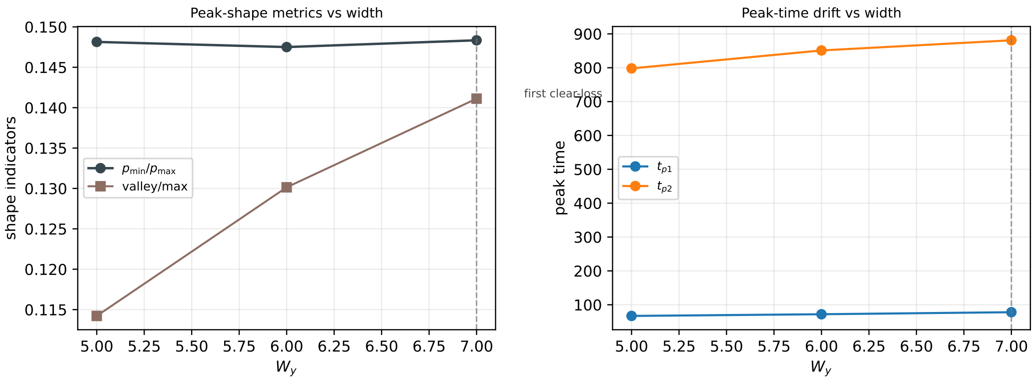

Figure 13: Width dependence of peak-shape metrics and peak-time drift.

the peak-time scales are reported in Table 5 (e.g., on branch $x_0 = 10$, $(t_{p1}, t_{p2}) = (72, 851)$ at $W_y = 6$ and $(78, 881)$ at $W_y = 7$).

3.5 Mathematical mechanism (splitting decomposition)

Treat the two targets as absorbing states and the rest as transient. With the block form

$$P = \begin{bmatrix} Q & R \\ 0 & I_2 \end{bmatrix}, \qquad R = [r_1 \; r_2],$$

the splitting first-hit pmfs are

$$f_1(t) = \alpha Q^{t-1} r_1, \qquad f_2(t) = \alpha Q^{t-1} r_2,$$

and therefore the total first-hit pmf satisfies the exact identity

$$F_{\text{any}}(t) = f_1(t) + f_2(t).$$

Double peaks appear when the conditional modes are separated and the splitting weights are not too extreme; we operationalize this via a separation score, valley depth, and a minimum weight threshold.

4 Study II: One Target With a Reflecting-Wall Corridor

4.1 Construction and scan variables

Start and target are placed near the two ends on the midline. Internal reflecting walls are added on a central x-span so that a corridor forms around the midline. We add an eastward local bias inside the corridor, and compare three regimes: $b_x = 0$ (no global drift), $b_x > 0$ (drift toward the target), and $b_x < 0$ (drift against the target), while scanning W_y. To avoid confounding width effects with arbitrary placement, we first run a small anchor pre-scan over candidate (x_s, x_t) and representative b_x values, then fix one anchor and scan widths. The selected anchor and the resulting clear interval are reported in Table 7 (script-generated to avoid manual transcription drift).

4.2 Phase maps and overview

From Table 6 and Table 7, under the selected anchor $(x_s, x_t) = (7, 58)$, **clear double** (phase 2) appears only at $b_x = -0.08$ with $W_y = 8, 10, 12, 14, 16$, then degrades to weak double (phase 1) from $W_y = 18$ onward. At $b_x = -0.12$, all scanned widths are weak-double (phase 1);

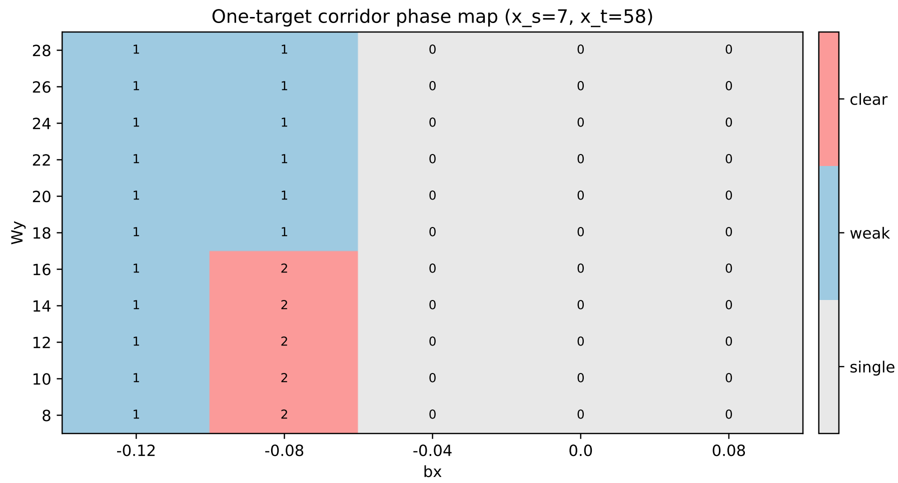

Figure 14: One-target corridor phase map on (W_y, b_x).

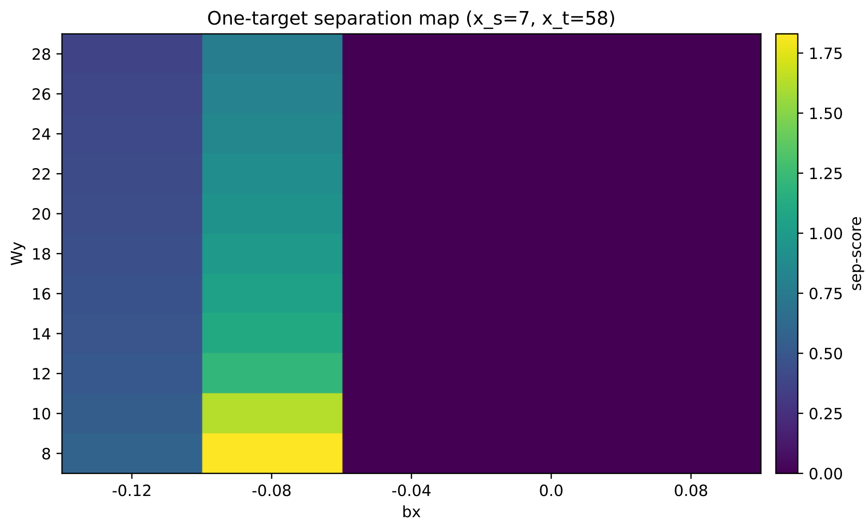

Figure 15: One-target separation score map.

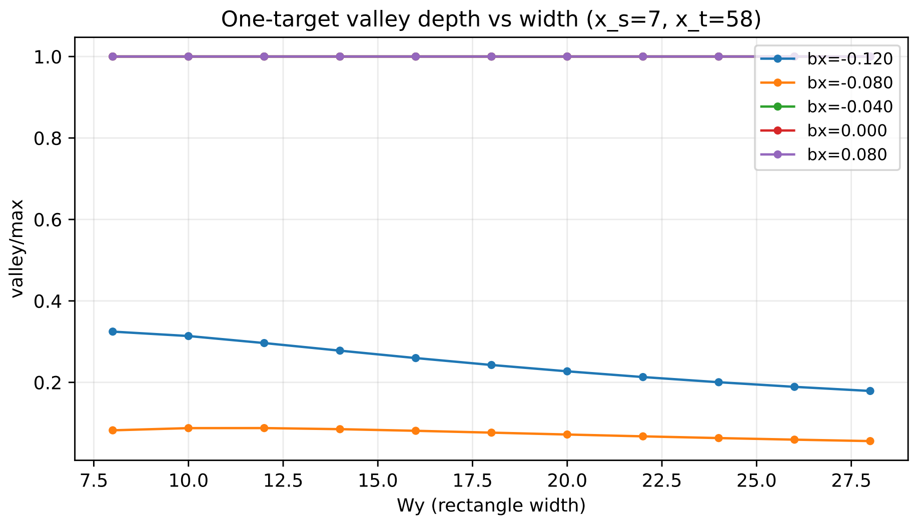

Figure 16: One-target valley depth vs width (multiple b_x curves).

W_y	$b_x = -0.12$	$b_x = -0.08$	$b_x = -0.04$	$b_x = 0.00$	$b_x = 0.08$
8	1	2	0	0	0
10	1	2	0	0	0
12	1	2	0	0	0
14	1	2	0	0	0
16	1	2	0	0	0
18	1	1	0	0	0
20	1	1	0	0	0
22	1	1	0	0	0
24	1	1	0	0	0
26	1	1	0	0	0
28	1	1	0	0	0

Table 6: One-target phase values per width and global bias.

x_s	x_t	b_x^*	W_y^{ref}	phase(ref)	clear interval(s)	first loss
7	58	-0.080	12	2	8-16 (step=2)	18

Table 7: One-target anchor pre-scan summary (the width scan is run after fixing this anchor).

at $b_x \in \{-0.04, 0, 0.08\}$, all scanned widths are single-peak (phase 0). So in this one-target corridor setup, a *moderate adverse* global drift is required for robust clear bimodality. Likewise, the reported critical point is a width threshold (for this anchor, first loss is at $W_y = 18$ for $b_x = -0.08$); corresponding peak-time scales are available in the scan tables/CSV outputs.

Note that Fig. 16 is a metric plot, not a geometry panel. Geometry differences are shown in Fig. 17: besides W_y, the key change is b_x (annotated with value and direction), which shifts the corridor/outer-channel balance and therefore changes peak strengths and peak-time separation in Fig. 18.

4.3 Representative configurations and channel decomposition

ID	W_y	b_x	phase	valley/max	sep-score
OT_W16_bxm0p080	16	-0.080	2	0.081	1.05
OT_W28_bxm0p080	28	-0.080	1	0.056	0.77
OT_W28_bxm0p120	28	-0.120	1	0.179	0.37
OT_W08_bx0p000	8	0.000	0	-	0.00

Table 8: Representative one-target metrics.

4.4 New: Corridor-thickness sensitivity (fixed anchor)

To directly test whether a one-cell corridor can produce clear bimodality, we run an extra scan at fixed anchor $(x_s, x_t) = (7, 58)$ over corridor half-widths $h \in \{0, 1, 2, 3\}$, i.e., actual band widths $2h + 1 = \{1, 3, 5, 7\}$ cells, while keeping other one-target settings unchanged.

h	band width	phase2 @ $b_x = -0.08$	first loss	phase≥ 1 @ $b_x = -0.12$	phase≥ 1 @ $b_x = +0.00$
0	1	-	-	8-28	-
1	3	8-16	18	8-28	-
2	5	8-26	28	8-28	-
3	7	10-28	-	10-28	-

Table 9: One-target corridor-thickness summary at fixed anchor (clear/weak intervals and first-loss width under selected biases).

Figures 22–23 and Table 9 show a clear trend: with a one-cell corridor ($h = 0$), the focus-bias branch is typically at most weak-double; thickening the corridor to 3/5/7 cells substantially

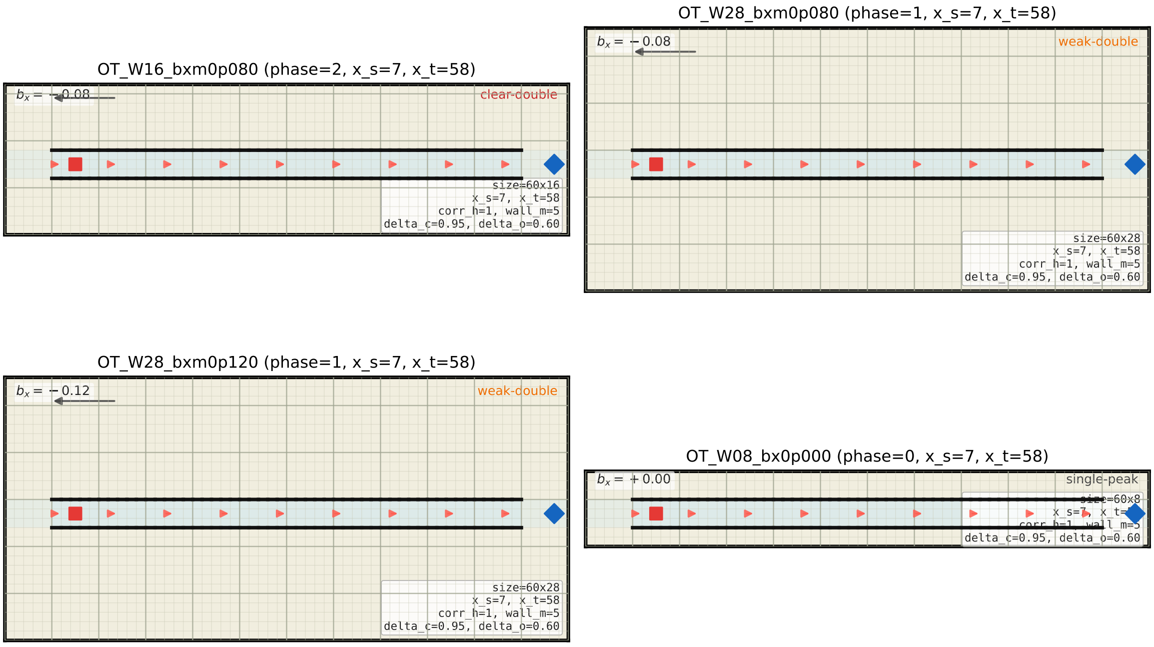

Figure 17: Representative corridor configurations (with reflecting walls). Each panel annotates b_x and its direction, plus a clear-double/weak-double/single-peak tag; a side config card reports size, (x_s, x_t), corridor half-width, wall margin, and $(\delta_{\text{core}}, \delta_{\text{open}})$.

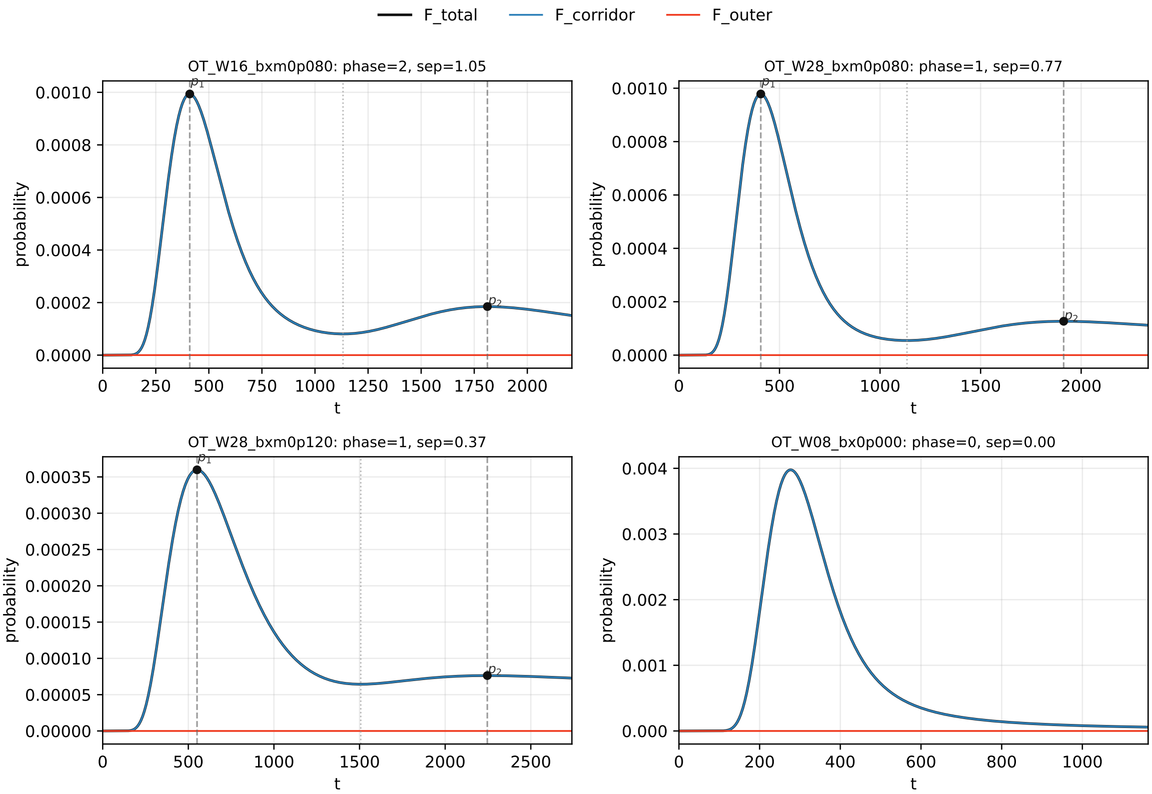

Figure 18: Representative one-target FPT curves with corridor/outer flux decomposition. For double-peak panels, both peaks p_1, p_2 are explicitly marked (dashed lines + black dots), with valley time marked when available. Each panel uses an adaptive x-window so the first lobe remains readable while still covering the second-peak timescale.

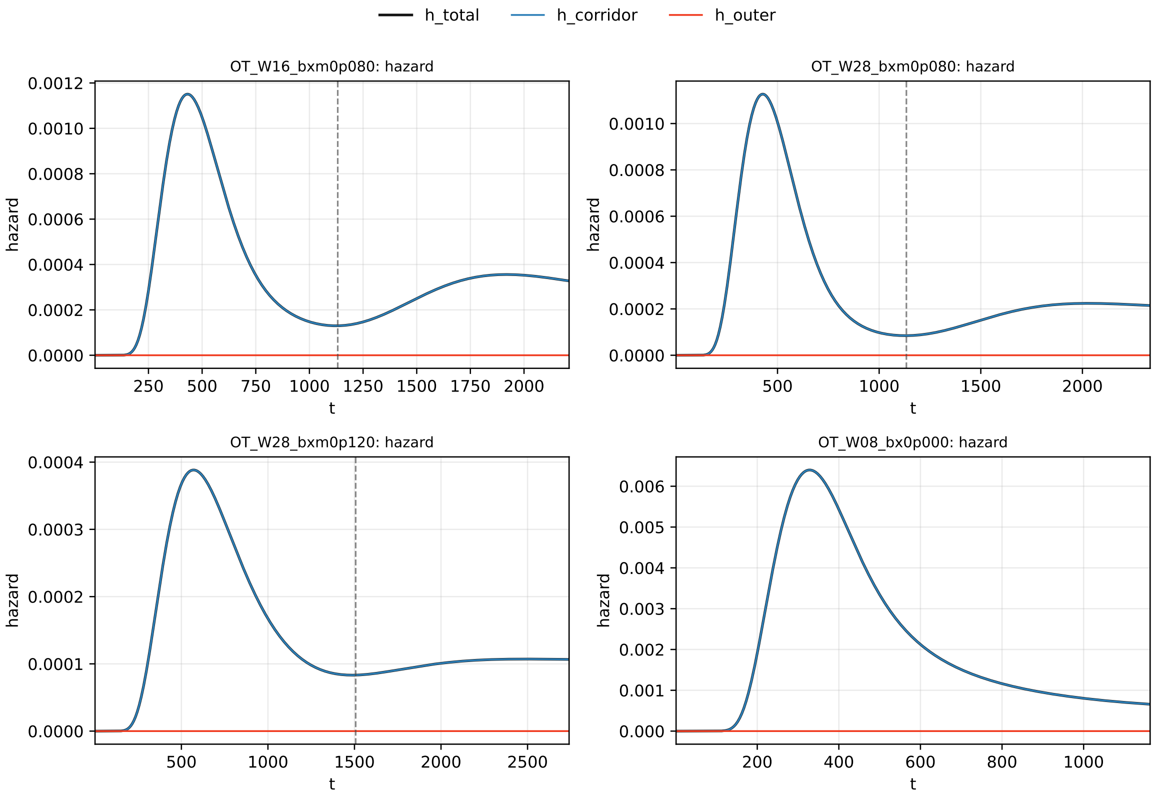

Figure 19: One-target hazard decomposition (corridor vs outer), with panel-wise adaptive x-windows aligned to the corresponding FPT panels to avoid long-tail compression of early-time structure.

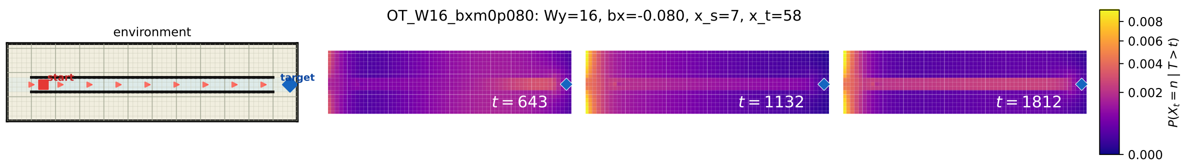

Figure 20: Representative case A: environment + conditional-occupancy heatmaps.

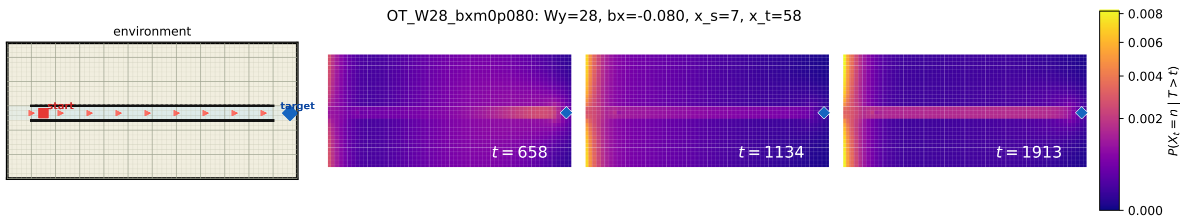

Figure 21: Representative case B: environment + conditional-occupancy heatmaps.

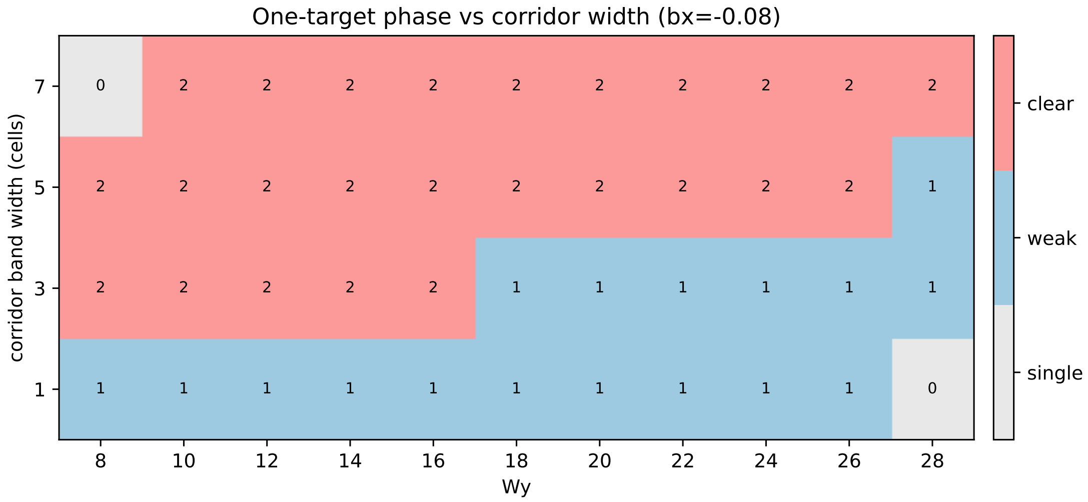

Figure 22: One-target phase over rectangle width W_y and corridor band width (in cells), at the focus bias.

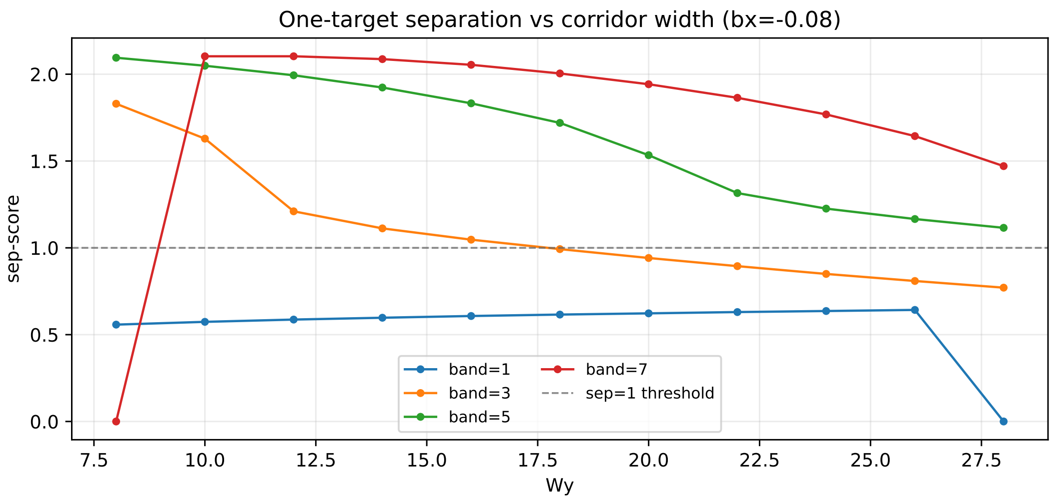

Figure 23: One-target separation score vs W_y for different corridor widths (dashed line: sep = 1 reference).

enlarges the clear-double regime. So an overly thin corridor weakens timescale separation, while moderate thickness improves robust bimodality.

4.5 New: Global-bias vector scan ((b_x, b_y))

At the same fixed anchor and a reference width (auto-selected as the nearest scanned W_y), we additionally scan the global-bias vector (b_x, b_y) to test how vertical drift modifies the one-target bimodality phase beyond the usual b_x sweep.

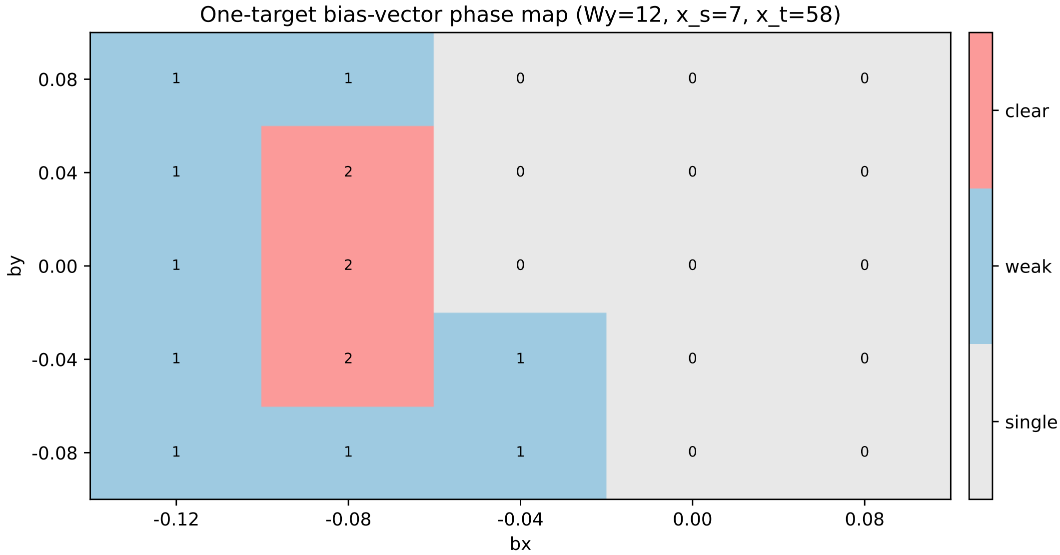

Figure 24: One-target phase map on (b_x, b_y) under fixed geometry.

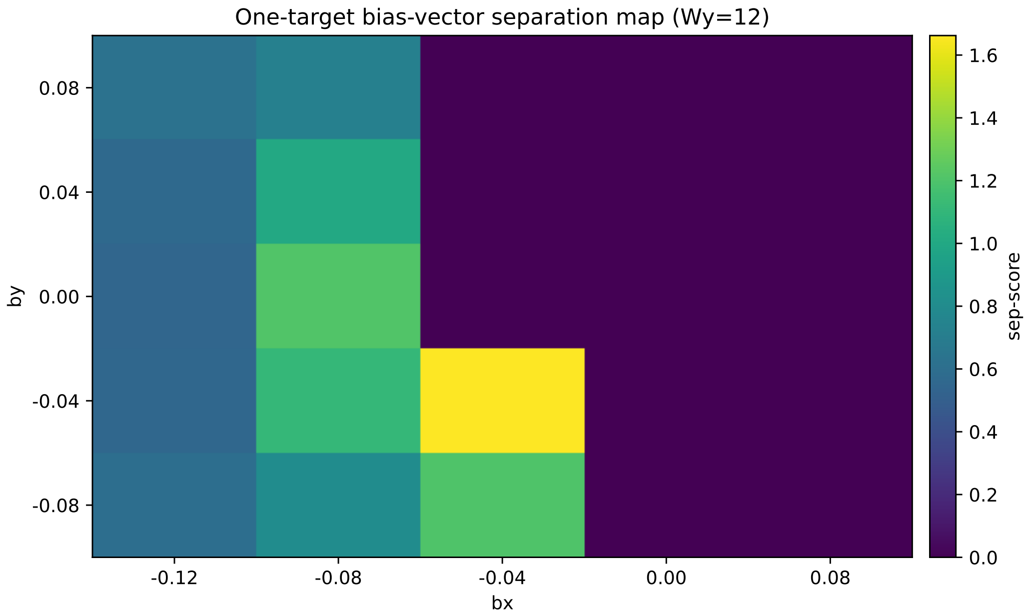

Figure 25: One-target separation-score map on (b_x, b_y) under fixed geometry.

Figures 24–25 and Table 10 provide a direct readout of how b_y changes the corridor-vs-outer channel balance, and therefore shifts double-peak robustness even at the same b_x.

b_y	$b_x = -0.12$	$b_x = -0.08$	$b_x = -0.04$	$b_x = 0.00$	$b_x = 0.08$
-0.08	1	1	1	0	0
-0.04	1	2	1	0	0
0.00	1	2	0	0	0
0.04	1	2	0	0	0
0.08	1	1	0	0	0

Table 10: Phase table for the (b_x, b_y) scan (matching Fig. 24).

4.6 Mathematical mechanism (two-channel absorption flux)

For one target, the first-hit pmf equals the total one-step flux into the absorbing state:

$$f(t) = \sum_{u \in \mathcal{T}} p_{t-1}(u)\, P(u \to m).$$

Partition $\mathcal{T} = \mathcal{C} \cup \mathcal{O}$ into corridor-band vs outer-band sources, yielding the exact identity

$$f(t) = f_{\mathcal{C}}(t) + f_{\mathcal{O}}(t), \qquad f_{\mathcal{C}}(t) = \sum_{u \in \mathcal{C}} p_{t-1}(u)\, P(u \to m).$$

When the two components have separated modes and comparable weights, their sum can become double-peaked (Fig. 18).

Reproducibility

```
# Install dependencies once at repo root
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

cd reports/grid2d_rect_bimodality
python3 code/rect_bimodality_report.py --quick
python3 code/rect_bimodality_report.py

latexmk -xelatex -interaction=nonstopmode -halt-on-error \
  -auxdir=build -emulate-aux-dir grid2d_rect_bimodality_cn.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -auxdir=build -emulate-aux-dir grid2d_rect_bimodality_en.tex
```